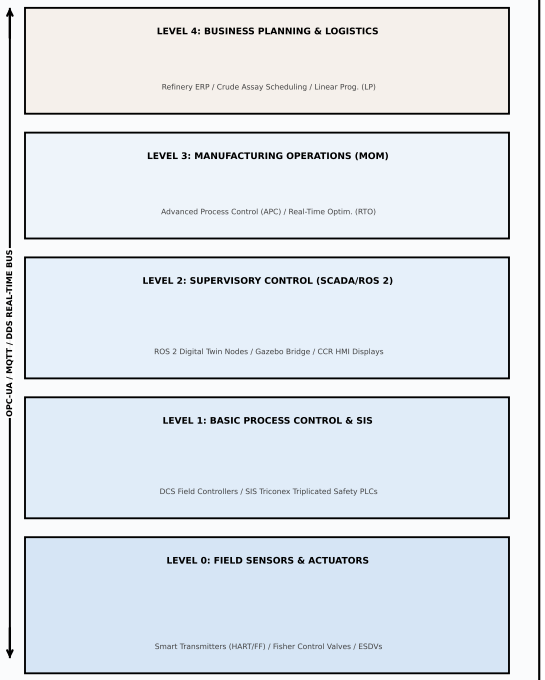
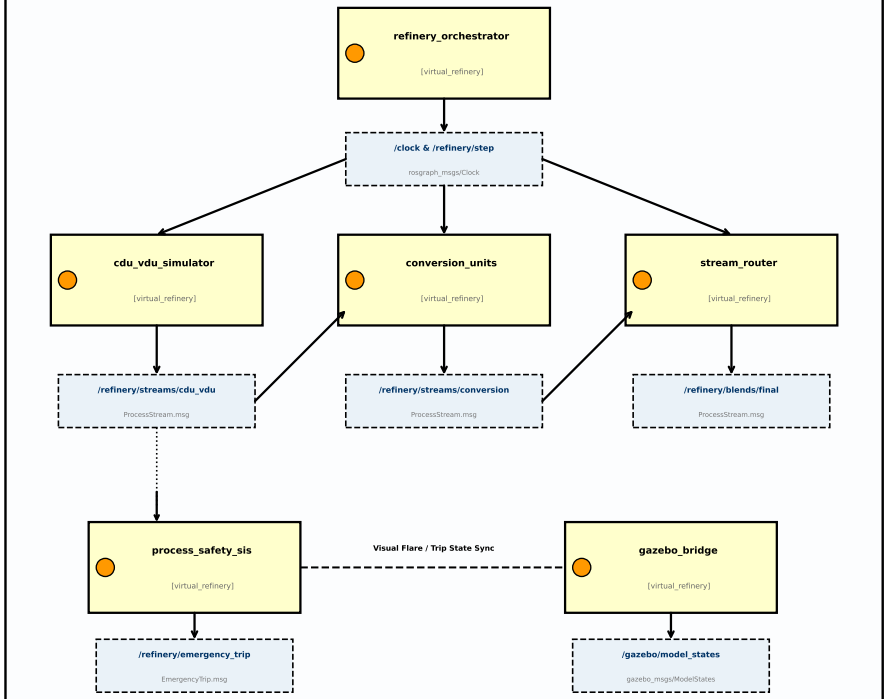
ROS 2 DIGITAL TWIN ARCHITECTURE, COMPUTATIONAL GRAPH & DCS INTEGRATION

ISA-95 CONTROL PYRAMID | ROS 2 HUMBLE/IRON PUB-SUB GRAPH | DDS MIDDLEWARE | OPC-UA GATEWAY

ISA-95 REFINERY AUTOMATION PYRAMID



ROS 2 REFINERY COMPUTATIONAL GRAPH & TOPIC MESSAGING



ROS 2 CUSTOM INTERFACES (.msg / .srv)

ProcessStream.msg:

```
std_msgs/Header header
string stream_name
float64 mass_flow_kg_h
float64 temperature_c
float64 pressure_bar
float64 enthalpy_kj_kg
string phase  # VAPOR / LIQUID
float64[12] component_fractions
```

UnitTelemetry.msg:

```
string unit_id  # CDU / VDU / FCC
float64 operating_pressure_bar
float64[] tray_temperatures_c
float64 liquid_level_pct
float64 differential_press_mbar
bool trip_interlock_active
```

SetCutPoints.srv:

```
float64 cdu_naphtha_cut_c
float64 cdu_kerosene_cut_c
float64 cdu_diesel_cut_c
---
bool success
string message
```

INDUSTRIAL AUTOMATION ARCHITECTURE, PROTOCOL STANDARDS & ROS 2 MIDDLEWARE INTEGRATION

AUTOMATION BUS / INTERFACE	PHYSICAL / NETWORK LAYER	CYCLE TIME / FREQ	SECURITY / PROTOCOL	PRIMARY FUNCTION IN REFINERY TWIN
ROS 2 FastDDS / CycloneDDS	Gigabit Ethernet (TCP/UDP)	1000 Hz / 1 ms	DDS-Security (TLS/PKI)	Inter-node process telemetry & unit coordination
OPC-UA Server (IEC 62541)	TCP/IP over Ethernet 802.3	10 Hz / 100 ms	X.509 Certificate Encrypt	SCADA / DCS HMI real-time data exchange
Modbus TCP / Fieldbus	RS-485 / Industrial Ethernet	20 Hz / 50 ms	Role-Based Access (RBAC)	Smart actuator commands & valve position feedback
HART 7.0 Digital Protocol	FSK superimposed 4-20 mA	2 - 4 Hz / 500 ms	Point-to-Point / Wireless	Transmitter calibration, diagnostics & PV polling
Safety Instrumented Net (SIS)	Dual Redundant Fiber Optic	< 10 ms Deterministic	TÜV Rheinland IEC 61508	Emergency shutdown interlocks (ESD-101 to ESD-401)
Gazebo Transport Engine	ZeroMQ / Protobuf Socket	1000 Hz Physics Step	Local IPC / Socket Domain	Synchronous rigid-body physics & visual mesh updates
Alarm & Event Logger (A&E)	PostgreSQL / InfluxDB TSDB	Event Driven (<5 ms)	AES-256 at Rest	Trip sequence of events (SOE) timestamp logging

JADAVPUR UNIVERSITY | CHEMICAL ENGINEERING DEPARTMENT | PROCESS PLANT DIGITAL TWIN SIMULATION

DRAWING TITLE	LAB / SUBJECT	COURSE / BRANCH	SCALE
ROS 2 DIGITAL TWIN & INDUSTRIAL AUTOMATION ARCHITECTURE	PROCESS PLANT DIGITAL TWIN & CAD	B.E. CHEMICAL ENGINEERING	LOGICAL / ARCHITECTURAL
STUDENT NAME	**ROLL NO.**	**DRAWING NO.**	**DATE**
SOURISH SENAPATI	002310301035	JU/ChE/VP/DWG-05	19 / 10 / 2026

VIRTUAL PETROLEUM REFINERY | 100,000 BPD CRUDE DIET | GAZEBO & ROS 2 DIGITAL TWIN | API / ASME BPVC